

Models

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R Markdown

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When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
#set up libraries  
library(caret)
```

```
## Loading required package: ggplot2
```

```
## Loading required package: lattice
```

```
library(ggplot2)  
library('naivebayes')
```

```
## Warning: package 'naivebayes' was built under R version 4.3.2
```

```
## naivebayes 0.9.7 loaded
```

```
library('C50')
```

```
## Warning: package 'C50' was built under R version 4.3.2
```

```
library(Metrics)
```

```
## Warning: package 'Metrics' was built under R version 4.3.2
```

```
##
```

```
## Attaching package: 'Metrics'
```

```
## The following objects are masked from 'package:caret':
```

```
##
```

```
##      precision, recall
```

```
library('tinytex')
```

```
## Warning: package 'tinytex' was built under R version 4.3.2
```

```
tinytex::install_tinytex(force = TRUE)
tinytex::is_tinytex()
```

```
## [1] TRUE
```

```
#Open file
getwd()
```

```
## [1] "C:/Users/Grant/OneDrive/Documents/SCHOOL/DTSC3010"
```

```
setwd("C:/Users/Grant/OneDrive/Documents/RFiles")
nbadf <- read.csv('NBAPlayerData.csv')
```

```
##create training and testing datasets
```

```
sample_size <- floor(0.8*nrow(nbadf))
#check size
sample_size
```

```
## [1] 7728
```

```
# Identify constant variables
constant_vars <- sapply(nbadf, function(x) length(unique(x)) == 1)

# Exclude constant variables
nbadf <- nbadf[, !constant_vars]

# get train data index
train_ind <- sample(seq_len(nrow(nbadf)), size = sample_size)
# generate training and test dataset
train <- nbadf[train_ind,]
test <- nbadf[-train_ind,]

#remove empty values
train <- train[complete.cases(train),]
test <- test[complete.cases(test),]

#set cross validation method: ## 5-fold CV , repeated two times
fitControl <- trainControl(method = "repeatedcv", number = 5, repeats = 2)
```

model #1: gaussian kernel

```
## svm with gaussian kernel
svmradiial <- train(MP ~ X3P.+eFG.+TRB+AST+STL+BLK+PTS, data = train,
  method = "svmRadial",
  trControl = fitControl)
#apply model and evaluate prediction results for svm gaussian
prediction_svmgaussian <- predict(svmradiial, newdata = test)
rmse(prediction_svmgaussian, test$MP)
```

```
## [1] 2.613321
```

model #2: Linear Regression

```
lm_model <- lm(MP ~ X3P. + eFG. + TRB + AST + STL + BLK + PTS, data = train)
```

```
# Extract summary  
summary(lm_model)
```

```
##  
## Call:  
## lm(formula = MP ~ X3P. + eFG. + TRB + AST + STL + BLK + PTS,  
##     data = train)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -21.4610  -2.2007  -0.0982   2.2385  20.9787   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)  4.568336   0.191227   23.890 < 2e-16 ***  
## X3P.         4.377687   0.286529   15.278 < 2e-16 ***  
## eFG.        -0.003949   0.422322   -0.009 0.992540   
## TRB         1.013870   0.029572   34.285 < 2e-16 ***  
## AST         0.640367   0.036240   17.670 < 2e-16 ***  
## STL         5.156319   0.144977   35.567 < 2e-16 ***  
## BLK         0.448414   0.130858    3.427 0.000614 ***  
## PTS         0.761244   0.012526   60.775 < 2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 3.516 on 7720 degrees of freedom  
## Multiple R-squared:  0.8715, Adjusted R-squared:  0.8714   
## F-statistic: 7482 on 7 and 7720 DF,  p-value: < 2.2e-16
```

```
##convert continuous minutes played values to labels by using a 75 percent threshold
```

```
threshold <- quantile(nbadf$MP, 0.75)  
nbadf$MP <- factor(with(nbadf, ifelse(nbadf$MP <= threshold, 'low', 'high')))
```

```
# now check number of smaples in each category  
table(nbadf$MP)
```

```
##  
## high low  
## 2407 7253
```

```
##create training and testing datasets
```

```
sample_size <- floor(0.8*nrow(nbadf))  
#check size  
sample_size
```

```
## [1] 7728
```

```

# Identify constant variables
constant_vars <- sapply(nbadf, function(x) length(unique(x)) == 1)

# Exclude constant variables
nbadf <- nbadf[, !constant_vars]

# get train data index
train_ind <- sample(seq_len(nrow(nbadf)), size = sample_size)
# generate training and test dataset
train <- nbadf[train_ind,]
test <- nbadf[-train_ind,]

#remove empty values
train <- train[complete.cases(train),]
test <- test[complete.cases(test),]

#set cross validation method: ## 5-fold CV , repeated two times
fitControl <- trainControl(method = "repeatedcv", number = 5, repeats = 2)

```

##Model #3: Decision tree

```

# Decision tree
svmradial2 <- train(MP ~ X3P.+eFG.+TRB+AST+STL+BLK+PTS, data = train,
                    method = "C5.0", trControl = fitControl)
#apply model and evaluate prediction results for decision tree
prediction_dectree <- predict(svmradial2, newdata = test)
confusionMatrix(prediction_dectree,test$MP)

```

```

## Confusion Matrix and Statistics
##
##           Reference
## Prediction high  low
##           high  446   68
##           low   69 1349
##
##               Accuracy : 0.9291
##               95% CI : (0.9167, 0.9401)
##           No Information Rate : 0.7334
##           P-Value [Acc > NIR] : <2e-16
##
##               Kappa : 0.8185
##
##  Mcnemar's Test P-Value : 1
##
##               Sensitivity : 0.8660
##               Specificity : 0.9520
##           Pos Pred Value : 0.8677
##           Neg Pred Value : 0.9513
##               Prevalence : 0.2666
##           Detection Rate : 0.2308
##   Detection Prevalence : 0.2660
##           Balanced Accuracy : 0.9090
##

```

```
##      'Positive' Class : high
##
```